

MACTAN CEBU INTL, Philippines

WMO: 986460

Lat: 10.3224N

Lon: 123.9800E

Elev: 31

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.1	75.0	67.8	102.3	79.8	69.5	108.4	80.1	22.6	81.5	20.6	81.2	5.5	0	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.8	91.7	80.6	91.1	80.5	89.9	80.3	83.6	88.4	82.8	87.7	82.3	87.3	9.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.5	169.4	86.8	81.7	164.9	86.4	80.9	160.5	86.0	47.7	88.4	46.9	87.8	46.3	87.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.6	16.3	14.3		70.2	94.4	3.4	1.5	67.8	95.4	65.8	96.3	63.9	97.1	61.5	98.2	
				DB	67.5	84.6	2.6	1.3	65.6	85.6	64.0	86.4	62.5	87.1	60.6	88.1
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.6	80.3	80.7	82.0	83.8	84.7	83.8	82.8	83.3	82.9	82.6	82.5	81.3	(6)
(7)		DBStd	2.08	1.58	1.57	1.59	1.58	1.70	1.99	1.70	1.75	1.72	1.55	1.51	1.65	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11884	938	860	992	1014	1077	1013	1016	1031	987	1011	974	971	(10)
(11)		CDD65	6409	473	440	527	564	612	563	551	566	537	546	524	506	(11)
(12)		CDH74	72159	4419	4323	5826	6987	7845	6785	6233	6503	6008	6120	5884	5226	(12)
(13)	CDH80	25096	962	1091	1880	2854	3508	2731	2186	2396	2106	2083	1916	1383	(13)	
(14)	Wind	WSAvg	6.5	8.3	8.2	7.7	6.8	5.6	5.1	5.8	6.3	5.7	5.5	6.3	7.2	(14)
(15)	Precipitation	PrecAvg	69.3	5.0	3.5	2.6	2.9	5.5	7.3	8.1	6.7	7.4	8.5	5.9	6.1	(15)
(16)		PrecMax	101.3	17.4	10.1	6.7	10.2	15.4	12.8	14.5	11.0	15.4	15.7	18.6	16.1	(16)
(17)		PrecMin	46.8	0.1	0.4	0.2	0.9	1.1	3.4	2.0	1.7	2.7	2.7	1.3	0.9	(17)
(18)		PrecStd	13.4	3.8	2.6	2.1	2.4	3.2	2.5	3.3	2.4	3.1	2.8	3.9	3.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.8	88.2	89.9	91.7	93.7	93.3	91.4	91.8	91.6	90.7	89.8	88.8	(19)
(20)			MCWB	79.2	79.5	78.6	79.6	81.3	81.5	80.9	80.6	80.9	81.2	80.2	80.1	(20)
(21)		2%	DB	86.3	87.3	88.6	90.3	91.7	91.5	89.8	90.4	90.0	89.5	88.4	87.7	(21)
(22)			MCWB	78.7	78.8	78.8	79.6	80.5	80.8	80.6	80.5	80.6	80.5	80.3	79.9	(22)
(23)		5%	DB	85.7	86.2	87.8	89.6	91.0	89.9	88.8	89.5	89.3	88.2	87.8	86.4	(23)
(24)			MCWB	78.5	78.3	78.7	79.5	80.4	80.9	80.5	80.3	80.5	80.5	80.3	79.5	(24)
(25)		10%	DB	84.5	85.4	86.8	88.6	89.7	88.9	87.7	88.1	87.9	87.5	87.0	85.8	(25)
(26)			MCWB	78.1	77.9	78.5	79.5	80.4	80.8	80.3	80.3	80.4	80.4	80.0	79.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	82.1	82.5	83.0	84.3	84.3	83.6	83.5	83.6	83.8	83.5	82.8	(27)
(28)			MCD B	84.8	85.8	86.4	87.5	89.9	90.0	88.6	88.2	88.3	87.8	87.1	86.0	(28)
(29)		2%	WB	80.6	80.7	81.4	82.1	83.0	83.0	82.5	82.3	82.5	82.6	82.3	81.6	(29)
(30)			MCD B	84.4	84.9	85.9	87.3	88.6	88.5	87.5	87.6	87.6	87.0	86.4	85.4	(30)
(31)		5%	WB	79.7	79.7	80.4	81.3	82.3	82.1	81.6	81.6	81.6	81.6	81.3	80.6	(31)
(32)			MCD B	83.7	84.2	85.5	86.9	87.9	87.6	86.7	87.0	86.9	86.4	85.9	84.8	(32)
(33)		10%	WB	78.9	78.9	79.7	80.6	81.6	81.3	80.8	80.8	80.8	80.8	80.5	79.8	(33)
(34)			MCD B	83.1	83.6	84.7	86.6	87.4	86.9	86.1	86.5	86.3	85.9	85.3	84.0	(34)
(35)	Mean Daily Temperature Range	MDBR	8.6	9.3	10.1	10.3	10.8	10.6	10.2	10.5	10.4	10.2	9.2	8.7	(35)	
(36)		5% DB	MCD BR	10.0	10.3	11.1	11.2	12.0	11.9	11.4	11.8	11.9	11.2	10.2	9.8	(36)
(37)			MCWBR	5.3	5.2	4.9	4.8	4.9	5.1	5.4	5.7	5.5	5.0	4.9	(37)	
(38)		5% WB	MCD BR	9.2	9.7	10.1	10.2	11.0	11.1	10.6	11.0	11.0	10.5	9.7	9.2	(38)
(39)			MCWBR	5.6	5.8	5.4	5.3	5.6	5.7	5.8	6.1	6.2	5.9	5.5	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.436	0.423	0.423	0.428	0.427	0.421	0.423	0.441	0.437	0.450	0.433	0.433	(40)	
(41)		taud	2.411	2.438	2.445	2.429	2.449	2.476	2.470	2.405	2.413	2.373	2.429	2.424	(41)	
(42)		Ebn at Noon	277	284	285	280	275	273	274	273	277	273	276	275	(42)	
(43)		Edh at Noon	37	37	37	38	36	35	35	38	38	39	36	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1268	1533	1787	1961	1813	1636	1545	1604	1585	1531	1435	1281	(44)	
(45)		RadStd	165	158	172	99	123	98	96	104	136	95	121	116	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.43	+0.38	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page